Renesas EK-RA8D1

Three axes, a real tachometer, and what speed does *not* buy

Core	Cortex-M85 480 MHz, Helium MVE
SRAM	**1 MB** — the real differentiator
Pipeline	1.3 ms per axis, 5.9 ms for three
Tacho	hardware capture — the only kit with one
Compute share	8.8 % of the energy cycle
Battery life	3.12 y (18650, self-discharge bound)

Never flashed. No Renesas toolchain in this environment, so the driver is compiled against a *stub* FSP under the real warning set — which proves its types and signatures, and not one register value.

The kit's distinctive module, angular resampling, **is** measured: 14 checks on a synthetic swept-speed signal. The measurement contradicted the design's own assumption, which is this deck's main finding.

Branch `hw/renesas-ra8d1-helium`. 19 config property tests, 14 order-tracking checks, 4 three-axis end-to-end tests, 1 113 shared DSP checks.

A faster core does not save energy

	Clock	Pipeline	Current	Energy
STM32U585 (M33)	160 MHz	8.6 ms	21 mA	0.18 mA·s
Alif E7 (M55 HE + Helium)	160 MHz	2.4 ms	9 mA	0.02 mA·s
RA8D1 (M85 + Helium)	**480 MHz**	**1.3 ms**	**62 mA**	**0.08 mA·s**
Acquisition window		**307 ms**	**6 mA**	**1.84 mA·s**

Energy is work and the work is fixed. The M85 is $6.6\times$ faster than the M33 and draws $3\times$ the current: a $2\times$ win on a term that is 9 % of the cycle.

If the only argument for a EUR 100 M85 is speed, the EUR 90 STM32U5 wins on price and the comparison is over.

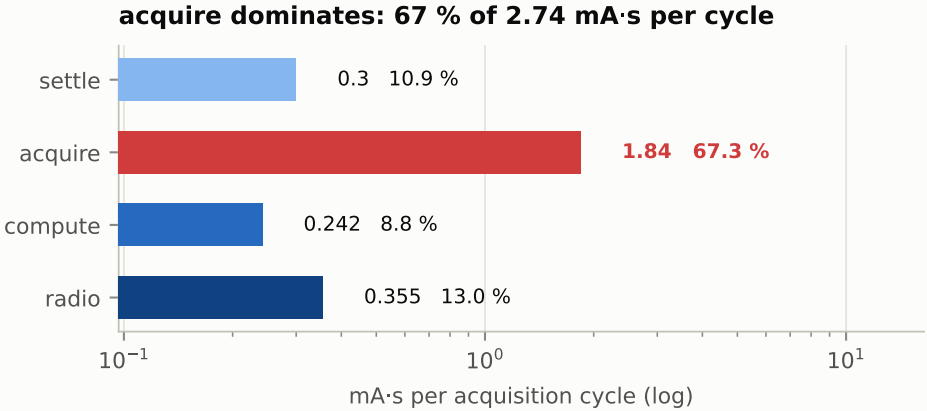
So the kit is configured to spend the headroom on **capability**, not on finishing early:

three live axes · a hardware tachometer · angular resampling · 227 kB of 1 MB used, with room for a 16384-point FFT or several bearings per node.

Total cost of all of it: **10 % of the cycle energy**.

So what: Buy this part for what it can hold and what it can reach, not for how fast it finishes.

Where the energy goes — and a correction

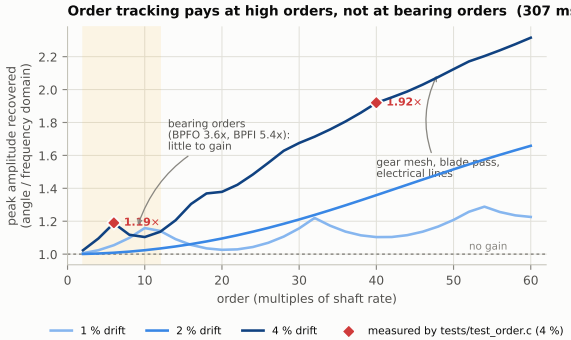


Acquisition dominates, as on every MEMS kit here. Compute is 8.8 % — three times a single-axis kit's share, and still small. Radio 13 %, also $3\times$: three frames per acquisition.

The model was wrong until this branch. It used a flat 1000 ms "standard acquisition" for every kit while the firmware acquires 8192 samples at 26.667 kHz = 307 ms — a $3.3\times$ error in the largest term, already quoted in four pushed READMEs.

Now derived from `acq_samples / fs_hz`, with a property test asserting it matches `app_config.h`.

Order tracking: the design's assumption was wrong



Smearing goes as (order × drift × record length). At a 307 ms record the bins are 3.26 Hz and a 6x line at 4 % drift smears over ∼7 Hz — barely wider than a Hann main lobe.

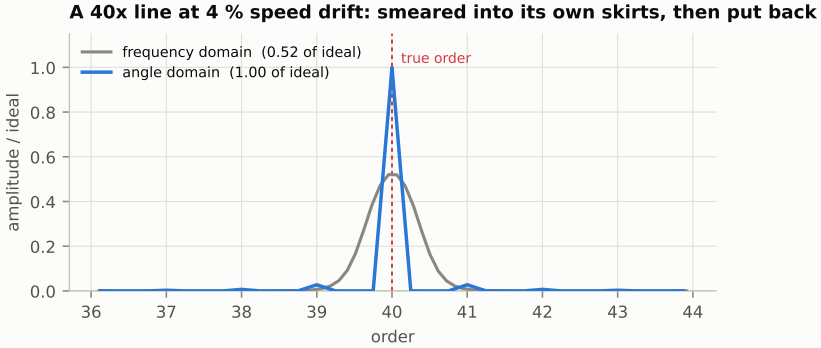
So resampling is nearly pointless at bearing orders here. The module's own docstring claimed 8–10 dB. The test said 1.5 dB.

It pays where an analyst actually fights: gear mesh, blade pass, electrical lines. At order 40 the same drift costs 5.67 dB and resampling returns $1.92\times$.

The red markers are the C test's measured values, parsed from its own output by the figure script — so the figure cannot quietly drift away from the test it illustrates.

`hardware/ra8d1/tests/test_order.c` — 14 checks.

The same line, both domains



A 40x line at 4 % drift: 0.52 of ideal amplitude spread over six bins, restored to 1.00 in one bin.

A trend built on the smeared peak tracks the VFD's duty cycle rather than the machine's condition — a false alarm every time production changes speed.

An unlooked-for result. On a *perfectly steady* shaft the frequency domain still loses 1.16 dB, because the line falls between bins. Scallop loss.

The angle domain puts it exactly on an order bin. For amplitude-based trending — most of ISO 20816 practice — that is arguably worth more than the smearing recovery.

Three axes on one wire format

One FIFO burst, three pipeline runs, three frames. The channel lives in **flag bits 8–9**, which were previously always zero — so every single-axis kit's frames are byte-identical and the schema version does not move.

What the shared loop guarantees:

- one burst, so the axes are the *same instant*;
- sequence per *frame* — ingest deduplicates on (device, sequence), so a shared sequence would discard two of three;
- channel 0 alone drives the alarm state machine;
- an overrun costs *one* frame, not three.

Two bugs, both plausible-and-wrong.

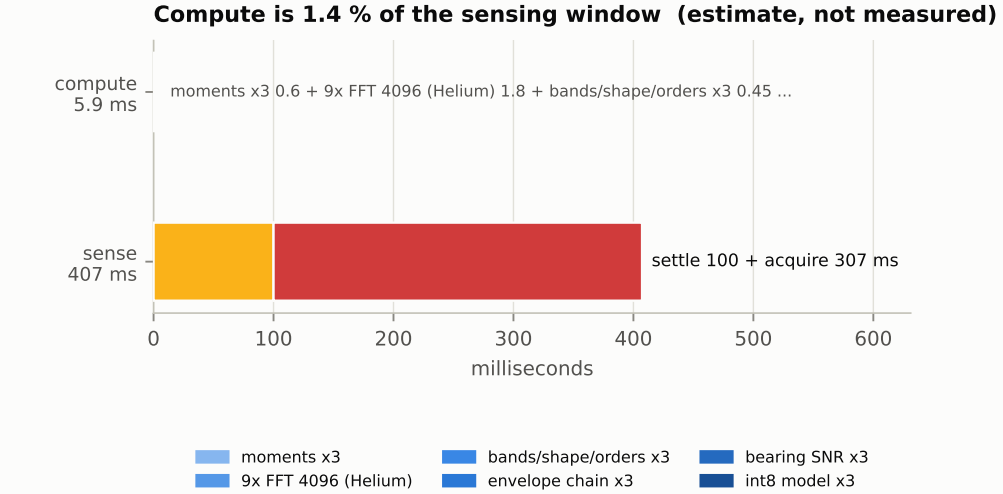
1. The shared loop powers the sensor down as soon as channel 0's burst is in RAM. Both boards checked "is the sensor powered?" *before* serving the cached channels, so axes 1 and 2 returned `NOT_READY` and the node shipped **one** frame per acquisition — correct features, correct flags, healthy-looking dashboard.

2. Power-down cleared the retained block, so the extra axes would have re-read the sensor: three different instants.

Both found on the first run of the three-axis end-to-end test.

So what: The multi-channel path went into the *shared* loop, guarded by `APP_N_AXES`, so single-axis builds emit identical bytes and 1 113 golden checks stay meaningful.

Compute is 1.4 % of the sensing window



5.9 ms for three axes plus a 0.9 ms resampling pass, against a 407 ms sensing window. There is room for a 16384-point FFT (1.63 Hz

The review item no test can catch.

`-mfpu=fpv5-sp-d16` compiles, links, produces bit-identical

What it does not do

The node does not ship order-domain features.

`pdm_pipeline_run` takes its sample rate from `app_config.h` at compile time. Feeding it a resampled record would label orders as hertz — numbers that look right and are wrong by whatever the speed happened to be.

So the node instead:

- flags `SPEED_UNSTEADY`, so the cloud widens its harmonic tolerance and discounts high orders;
- requests a raw snapshot, so the cloud does the angle-domain work;
- computes the local order spectrum anyway, for sites with no snapshot bandwidth.

It deliberately does **not** set `ORDER_TRACKED`: claiming the angle domain while shipping hertz is worse than no flag.

Shared DSP vs Python	1 113 checks
Resampling measured	14 checks
Three-axis path end-to-end	4 tests
Config properties	19 tests
Driver under `-Werror`	stub FSP
Compiled for M85	no
Flashed	no
Helium verified	no
Timing measured	no
Current measured	no
Tacho on a rotor	no

Three files are referenced by the build and not written. They are listed rather than stubbed, so the gap is a link error — not a function that returns success.

When to choose this part

Choose it for a VFD-driven critical asset where one radial channel is not enough: a gearbox where the mesh order matters, a fan where blade pass matters, a compressor whose duty varies through the day.

It costs more than the STWIN, draws more current, and needs wiring the STWIN does not. It buys an axial channel, a *measured* shaft speed, amplitude-accurate order analysis, and an order of magnitude of DSP headroom.

Do not choose it for a non-VFD motor-pump set with one accessible bearing. There the STWIN is the better engineering choice at two-thirds the price, with the sensor and radio already on the board, and this part is over-specified.

That is the honest recommendation, and the reason both branches exist.

So what: The tachometer, not the clock speed, is what makes this kit different — and the measurement showed it pays at gear orders, not at bearing orders.